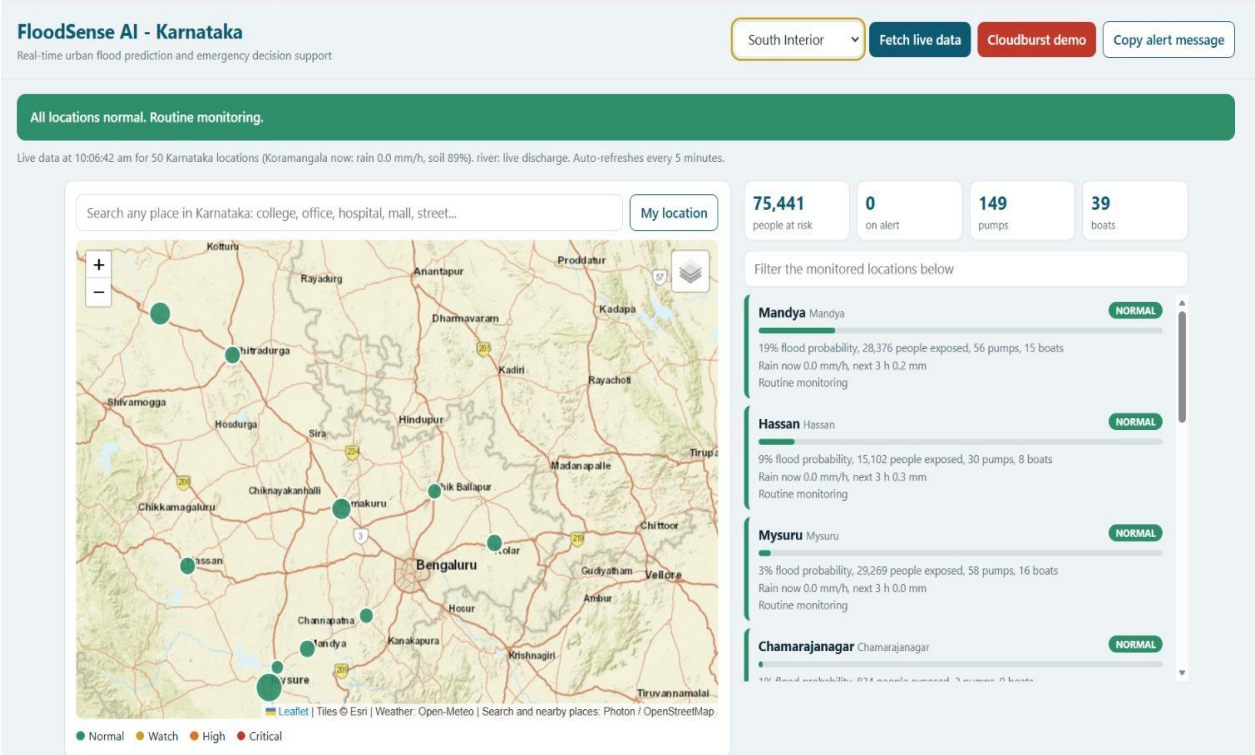
FloodSense AI – Karnataka

Real-time Urban Flood Prediction and Emergency Decision Support System



Live dashboard, 18 October 2026, 8:56 am

Project Report: Problem, Solution, System Design and Impact

Table of Contents

No.	Section	Page
1	Abstract	2
2	Introduction	3
3	Background: Floods in Karnataka	3
4	Problem Statement	4
5	Objectives	5
6	Proposed Solution	5
7	System Architecture	6
8	Data Sources and Inputs	6
9	Flood Prediction Method	7
10	Risk Levels and Recommended Actions	7
11	Dashboard Walkthrough (from the photo)	8
12	Case Study: Live Risk Locations	9
13	Cloudburst Simulation	9
14	Benefits, Users and Impact	10
15	Limitations and Future Scope	10
16	Conclusion	10

Abstract

Urban flooding has become one of the most damaging and frequent disasters in Indian cities. In Karnataka, and especially in Bengaluru, short bursts of intense rainfall regularly overwhelm drains and lakes, flooding roads, homes and offices within hours. The main reason these events cause so much damage is not only the rain itself, but the lack of an early, location-specific warning and a clear plan for where to send help.

FloodSense AI is a real-time web-based decision support system that monitors 50 flood-prone locations across Karnataka. It continuously gathers live weather data - current rainfall, short-term rain forecast, soil moisture and river discharge - and combines it with local terrain knowledge to calculate a flood probability for every location. Each location is then classified as Normal, Watch, High or Critical, displayed on an interactive map, and paired with a concrete action such as sending SMS alerts, pre-positioning pumps or diverting traffic.

At the time of the captured screenshot, the system was tracking 4,03,417 people at risk, with 4 locations on alert and a plan requiring 146 pumps and 38 rescue boats. Bellandur (60%) and Silk Board / HSR Layout (50%) were flagged as HIGH risk. This report explains the problem, the solution, how the system works, and how it can help authorities and citizens act before a flood, not after.

1. Introduction

A flood is an overflow of water onto land that is normally dry. In cities this usually happens when rainwater cannot drain away fast enough. Concrete roads and buildings stop water from soaking into the ground, drains get blocked with waste, and lakes that once absorbed excess water have been encroached upon or filled in. As a result, even moderate rainfall can turn major junctions into rivers.

Traditional flood management in many cities is reactive: teams are mobilised after residents complain or after the news reports flooding. By then, vehicles are stranded, houses are damaged and people are in danger. Modern technology - live weather APIs, open maps and data-driven prediction - makes it possible to move from reaction to prevention. FloodSense AI is built on this idea.

2.Background: Floods in Karnataka

Karnataka faces two types of flood risk. Coastal and Malnad regions (Udupi, Dakshina Kannada, Kodagu, Shivamogga) receive extremely heavy monsoon rain that causes rivers to rise and landslides. North Karnataka districts along the Krishna, Bhima and Tungabhadra rivers face riverine floods when upstream dams release water. Bengaluru and other growing cities face urban flash floods caused by short, intense rainfall.

Why Bengaluru floods repeatedly

- **Loss of lakes and wetlands:** the city once had hundreds of interconnected lakes; many have shrunk or vanished, removing natural storage.
- **Encroached storm-water drains (rajakaluves):** buildings over drains reduce their capacity.
- **Rapid concrete growth:** more paved surface means more runoff and less absorption.
- **Low-lying hotspots:** areas like Bellandur, Silk Board, HSR Layout, Bommanahalli and Koramangala sit in valleys where water collects.
- **Cloudbursts:** very heavy rain in a short time (over 100 mm in a few hours) that no drain system is designed for.

Past events such as the September 2022 Bengaluru floods, when tech corridors on Outer Ring Road were submerged and residents had to be rescued by boats and tractors, showed clearly how costly the lack of early warning can be.

2. Problem Statement

How can city and state authorities know, in advance and for each specific area, where flooding is likely, how many people will be affected, and what resources must be sent - fast enough to act before damage occurs?

Today this question cannot be answered easily. The key problems are:

2.1 Data is scattered

Rainfall readings, weather forecasts, soil moisture, lake levels and river discharge come from different agencies and websites. Nobody sees them together, so the combined danger - for example, wet soil plus forecast rain - goes unnoticed.

2.2 No location-level early warning

General warnings like 'heavy rain expected in Bengaluru' are not useful for a ward officer. They need to know which locality, how severe and when.

2.3 Reactive emergency response

2.4 Poor resource allocation

Pumps and boats are limited. Without a risk ranking, they may be sent to less critical areas while high-risk zones wait.

2.5 Citizens are not informed

Commuters drive into flooded underpasses and junctions like Silk Board because no timely, local alert reaches them.

2.6 Consequences

Area	Impact
Life and safety	Drowning, electrocution, people trapped in homes and vehicles
Economy	Damaged property and vehicles, lost working hours, disrupted IT corridors
Traffic	Hours-long gridlock at junctions and arterial roads
Health	Water-borne diseases, contaminated drinking water
Governance	Public anger, emergency spending, loss of trust

3. Objectives

- Collect live weather and hydrological data automatically for every monitored location.
- Predict the flood probability for each location in near real time.
- Classify risk into simple, colour-coded levels anyone can understand.
- Estimate people exposed and the pumps and boats needed.
- Recommend clear actions to authorities for each risk level.

- Let citizens search any place and check its flood risk.
- Simulate extreme events (cloudbursts) to test preparedness.

4. Proposed Solution: FloodSense AI

FloodSense AI is a single web dashboard that brings data, prediction and decisions together. Instead of looking at five different sources, an official opens one screen and sees: where the risk is, how big it is, and what to do.

Key features

Feature	Description
Live data fetch	Pulls rain, forecast, soil moisture and river discharge for 50 locations; auto-refreshes every 5 minutes.
Flood probability	Each location gets a percentage score (e.g. Bellandur 60%).
Risk map	Interactive Leaflet map with colour-coded circles for each location.
Top alert banner	The most urgent areas and actions shown at the top of the screen.
Impact counters	People at risk, locations on alert, pumps and boats required.
Location cards	Per-area details: probability, people exposed, pumps, boats, rain now and next 3 h, action.
Search and My location	Check any college, office, hospital, mall or street in Karnataka.
Region filter	View all Karnataka or a specific region.
Cloudburst demo	Simulates extreme rainfall to show how the system reacts.

5. System Architecture

The system follows a simple five-layer pipeline. Each layer passes its output to the next:

Layer	Role	Output
1. Data layer	Calls live weather APIs (Open-Meteo) for each location's coordinates	Rain, forecast, soil moisture, river discharge
2. Processing layer	Cleans and normalises readings, combines with terrain risk	Feature set per location
3. Prediction layer	Computes flood probability score	0-100% probability
4. Decision layer	Maps probability to a risk level, estimates people and resources, chooses action	Level, exposure, pumps, boats, action
5. Presentation layer	Web dashboard: map, banner, counters, cards, search	What officials and citizens see

Flow: Live APIs -> Data processing -> Prediction model -> Risk and action engine -> Dashboard and alerts -> (repeat every 5 minutes)

6. Data Sources and Inputs

Input	Source	Why it matters
Rainfall now (mm/h)	Open-Meteo	Shows how much water is falling at this moment
Rain next 3 hours	Open-Meteo forecast	Gives lead time to act before water arrives
Soil moisture (%)	Open-Meteo	Saturated soil cannot absorb more water, so runoff rises
River discharge	Open-Meteo flood API	Indicates riverine flood danger
Location and terrain	Built-in location list	Low-lying and historically flooded areas get higher base risk
Population	Area estimates	Converts risk into people exposed
Maps and search	Esri tiles, OpenStreetMap, Photon	Base map and place search

Example from the screenshot: Koramangala had 0.0 mm/h rain but **soil at 90%**. This means the ground is already full, so even light rain in the next hours could cause waterlogging - something a rain-only warning would miss.

7. Flood Prediction Method

FloodSense AI estimates flood probability by combining several factors into one score. Conceptually:

Flood probability = f(rain now, rain forecast, soil moisture, river discharge, local terrain risk)

- **Rain now** and **rain forecast** carry the most weight, because they directly add water.
- **Soil moisture** acts as a multiplier: the wetter the ground, the more of the rain becomes runoff.
- **River discharge** raises risk for locations near rivers.
- **Terrain / history** gives each location a base risk - Bellandur and Silk Board start higher than elevated areas.
- The score is recalculated every 5 minutes so it always reflects current conditions.

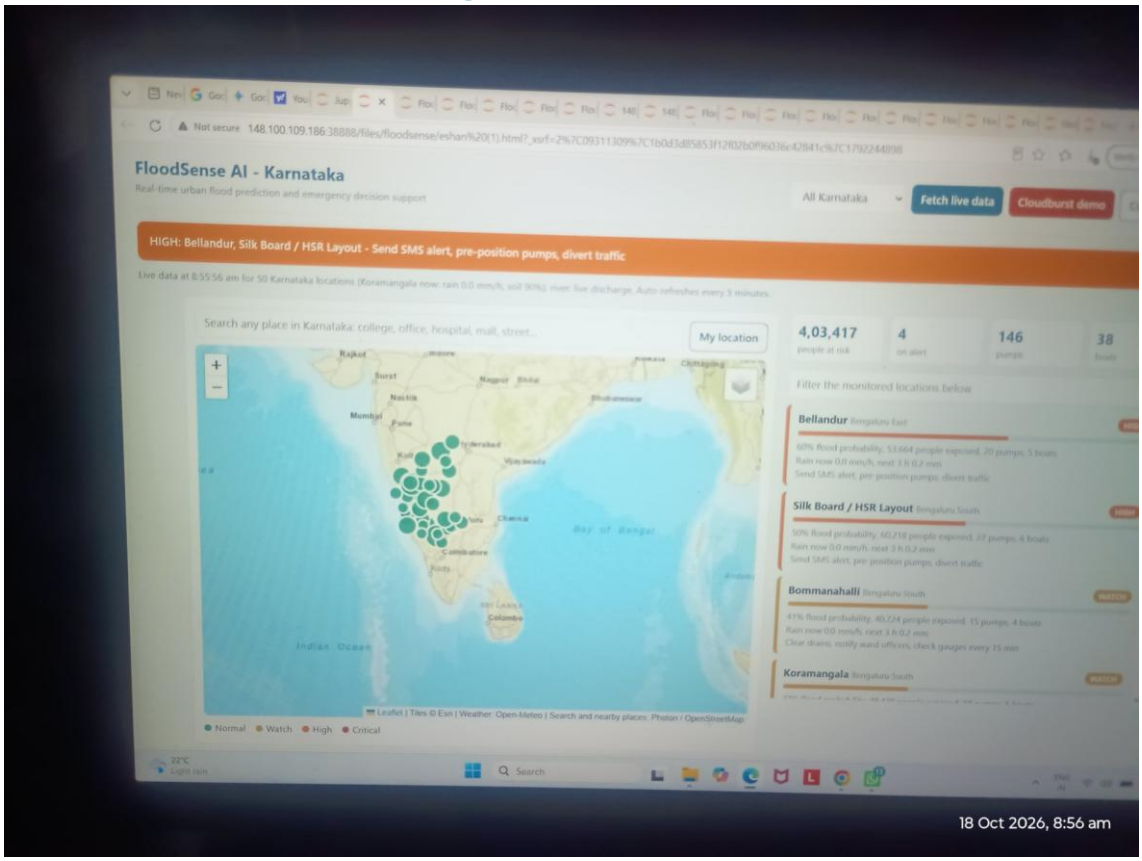
Resource estimation

Once the probability is known, the system multiplies it with each area's population to estimate **people exposed**, and uses the risk level and area size to estimate the **number of pumps and boats** to pre-position. For example, Silk Board / HSR Layout: 50% probability, 60,218 people exposed, 22 pumps and 6 boats.

8. Risk Levels and Recommended Actions

Level	Meaning	Recommended action
Normal	Low chance of flooding	Routine monitoring
Watch	Conditions building up	Clear drains, notify ward officers, check gauges every 15 minutes
High	Flooding likely soon	Send SMS alerts, pre-position pumps, divert traffic
Critical	Flooding happening or imminent	Deploy boats and rescue teams, evacuate low-lying homes, close roads

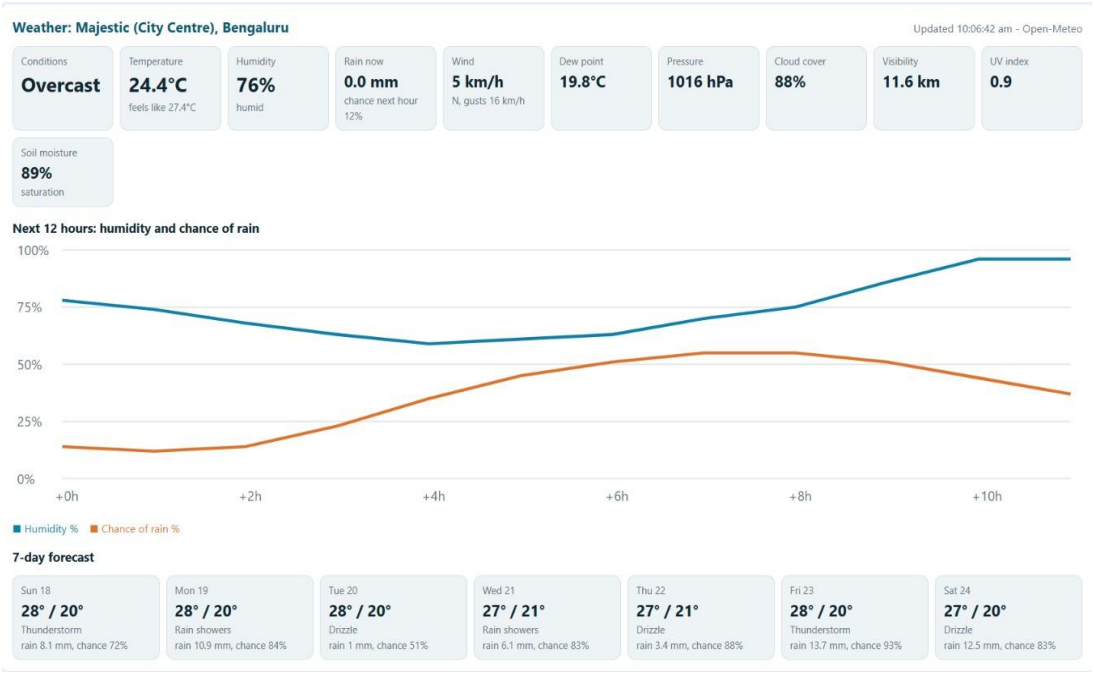
9. Dashboard Walkthrough

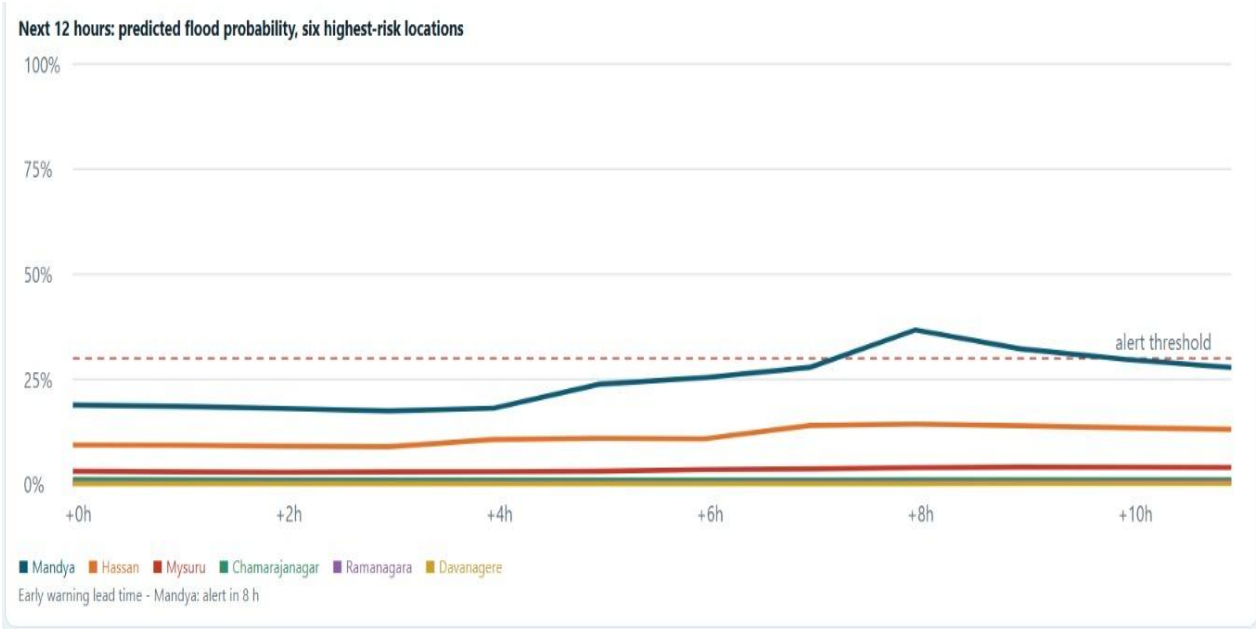


The FloodSense AI dashboard as captured

- **Header:** 'FloodSense AI - Karnataka' with the tagline 'Real-time urban flood prediction and emergency decision support'.
- **Controls (top right):** region selector 'All Karnataka', blue *Fetch live data* button, red *Cloudburst demo* button.
- **Alert banner:** 'HIGH: Bellandur, Silk Board / HSR Layout - Send SMS alert, pre-position pumps, divert traffic'.
- **Status line:** live data at 8:55:56 am for 50 locations, auto-refreshing every 5 minutes.
- **Search bar and My location:** lets anyone check risk for a specific place.
- **Map:** green circles across Karnataka show monitored locations; legend shows Normal, Watch, High, Critical.
- **Counters:** 4,03,417 people at risk | 4 on alert | 146 pumps | 38 boats.

- **Location cards:** ranked list with risk bar, details and action for each area.





10. Case Study: Live Risk Locations

Location	Zone	Level	Prob.	Exposed	Pumps	Boats
Bellandur	Bengaluru East	HIGH	60%	53,664	20	5
Silk Board / HSR	Bengaluru South	HIGH	50%	60,218	22	6
Bommanahalli	Bengaluru South	WATCH	41%	40,724	15	4
Koramangala	Bengaluru South	WATCH	-	-	-	-

Interpretation: rain at that moment was 0.0 mm/h with only 0.2 mm forecast in 3 hours, yet the system still flagged HIGH risk in Bellandur and Silk Board. This is because soil was heavily saturated from earlier rain and these areas are low-lying. The system warns officials that even a small additional shower could cause flooding, giving them time to prepare.

Decisions supported: together, the two HIGH locations alone account for over 1.1 lakh exposed people and need 42 pumps and 11 boats. Officials can send these resources there first, while Watch areas get drain clearing and closer monitoring.

11. Cloudburst Simulation

The *Cloudburst demo* button simulates a sudden extreme rainfall event across the monitored locations. This lets authorities:

- See how quickly locations move from Normal to High and Critical.
- Test whether enough pumps and boats are available in a worst case.
- Train staff on what actions to take before a real event.
- Show decision makers the value of early warning in a clear, visual way.

12. Benefits, Users and Impact

User	How FloodSense AI helps
City corporation (BBMP)	Prioritise pumps, drain clearing and road closures by ward
Disaster management (KSDMA)	State-wide view, resource planning, cloudburst drills
Traffic police	Divert traffic away from junctions before they flood
Citizens	Search their home, office or college and get alerts
Hospitals and schools	Plan closures and protect vulnerable people

- **Early action** instead of late reaction.
- **Lives saved** through timely alerts.
- **Lower losses** to homes, vehicles and businesses.
- **Efficient use** of limited pumps and boats.
- **Transparency**: one shared view for every department.

13. Limitations and Future Scope

- Accuracy depends on weather data quality; adding local rain gauges and lake-level sensors (IoT) would improve it.
- Train a machine learning model on historical flood records for more precise probabilities.
- Automatic SMS / WhatsApp alerts directly to residents of affected areas.
- Drain and lake capacity data from the city to model real drainage.
- Kannada language support and a mobile app for citizens. • Expand from 50 locations to every ward and other states.

14. Conclusion

Floods in Karnataka cannot always be stopped, but their damage can be greatly reduced with early, precise warnings. FloodSense AI turns scattered live data into clear, location-wise flood probabilities, impact estimates and actions on a single dashboard. As shown in the live screenshot, it can flag high-risk areas like Bellandur and Silk Board even before heavy rain arrives, and tell officials exactly how many people are exposed and what resources to send. This shift from reacting to preventing is the core value of FloodSense AI.